Problem Chosen	2026 HiMCM Summary Sheet	Team Control Number
A		18213

Choosing a City for High-School Life with Incomplete Evidence

Summary

Among these sixteen cities, Zurich scores highest under priorities a high schooler would set, with 64.5 points, a 1.2-point lead over Madrid; Singapore follows at 62.5. Zurich stays first in all 31 source-variant checks of counting rules and source choices for transit stops, pools and travel. In broader withdrawals, Madrid leads when 64 entries dated after 2025 are treated as unknown, and Milan when 28 flagged entries or their 95-entry union with the dated groups are unknown. Zurich also leads in 82.8% of 10,000 weight trials (concentration 100) and has the smallest maximum regret, 6.1 points: the most it could trail the best rival when every uncertain input breaks against it. The homicide input matters: withdrawing Zurich's homicide value (treating it as unknown, utility range $[0, 1]$) makes Madrid first.

We compare ten everyday measures using the analytic hierarchy process (AHP) to assign weights. Each measure becomes a utility, a score from 0 to 1. Safety and education each receive 34% of the category weight. The representative score is TOPSIS computed with the midpoint of each cell's utility range. Of 160 entries, 123 are single values, including proxies, and 37 are utility ranges. Nine cities could lead under favorable endpoints. Zurich remains first when education or mobility weights are doubled separately, and the three hypothetical 2035 scenarios keep the top three unchanged.

Keywords: high-school city choice; AHP; TOPSIS; interval evidence; maximum regret; student priorities.

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1 Introduction and decision scope

1.1 What makes a city suitable for high-school life?

A good city supports a student's school day through safety, affordable living, teaching resources and convenient travel. We compare sixteen cities, one per country, on ten everyday measures. Zurich leads Madrid by 1.2 points under our high-schooler priorities. Section 5.5 tests how source choices, uncertain inputs and different priorities affect that recommendation.

A utility puts each measure on a scale from 0 to 1, with higher values better. The **representative score** is TOPSIS computed with the midpoint of each cell's utility range.

1.2 Criterion selection and contribution

Our criteria cover safety, air quality, housing and living costs, teaching resources, higher education, transport stops, journey time, broadband and swimming facilities (Table 3). Friendships, language and cultural fit depend on the student; comparable surveys of wellbeing and discrimination were unavailable. City totals also omit fees, eligibility and neighbourhood routes. Families should check these factors against their needs and budget.

AHP gives pupils per teacher more weight than pools, reflecting their importance to daily schooling. Compared with equal weights, twelve cities change rank; Zurich remains first. The appended AI-use report describes assistance with research, calculations and writing [7, 8].

2 Assumptions and their role

1. **Source scopes stay fixed.** Mixed-year values retain their boundaries, populations and school stages, so comparisons describe those source periods.

2. **Utilities use fixed scales.** More resources and lower burdens improve utility; reference levels set the scales.

3. **Weights express our student priorities.** Safety and education each receive 0.340, cost and mobility 0.123 each, and daily life 0.073, without a survey.

4. **Strengths can offset weaknesses.** Families should apply essential safety or eligibility thresholds before using the combined score.

5. **Utility ranges vary independently.** Endpoint tests of the 37 utility ranges assume independent variation; possible winners and maximum regret depend on it.

6. **Scenarios hold the comparison basis fixed.** Population denominators, reference levels and weights stay unchanged.

3 Notation

Symbol	Meaning
x, a	Observed indicator value and reference level (in the same units)
u, u_{ij}	Utility on $[0, 1]$; i identifies a city and j an indicator
c, P, d	Count, population and density (institutions per million; stops and pools per 100,000)
t, q	Travel duration in minutes and its utility
w_j	Indicator weight, with all ten weights summing to one
D_i^+, D_i^-	City i's distances to the ideal and worst utility profiles
S_i	City i's TOPSIS score, on $[0, 100]$
ℓ_{ij}, h_{ij}	Lower and upper utility bounds
L_i, U_i	Lower and upper score bounds, $S(\ell_i)$ and $S(h_i)$
R_i	Maximum regret
κ	Concentration of a declared Dirichlet weight distribution
r	Raw-value multiplier in a development scenario

4 Indicator construction and city selection

4.1 City selection

The sixteen cities are Amsterdam, Berlin, Dublin, Helsinki, London, Madrid, Milan, Oslo, Paris, Prague, Singapore, Stockholm, Tokyo, Vienna, Warsaw and Zurich. The primary comparison uses one city from each of sixteen countries. The set spans cost, climate, population scale, mobility and safety (Table 2). Fourteen cities are European and all are in high-income economies; the recommendation applies to these sixteen candidates.

The locations and final ranks are shown in Figure 1.

One city per country avoids repeating national data systems and school structures. City records take priority; national statistics or surveys provide the labelled proxies listed in Section 4.3. Table 1 counts the evidence labels; some team-table values lack original links and are tested by withdrawal in Section 5.5.

Table 1: City profiles and evidence labels. Direct values are official or team-table values; ranges and proxies are counted separately.

City	Country	Population (year)	Direct values	Utility ranges	Proxies
Vienna	Austria	2,028,399 (2025)	8	1	1
Zurich	Switzerland	440,935 (2025)	6	3	1
Helsinki	Finland	694,392 (2025)	7	2	1
Berlin	Germany	3,700,577 (2025)	8	2	0
London	United Kingdom	9,122,909 (2025)	7	2	1
Amsterdam	Netherlands	941,927 (2025)	7	2	1
Stockholm	Sweden	999,200 (2025)	6	4	0
Paris	France	2,103,778 (2023)	7	2	1
Madrid	Spain	3,527,924 (2025)	7	2	1
Milan	Italy	1,399,079 (2025)	6	3	1
Dublin	Ireland	592,713 (2022)	7	2	1
Oslo	Norway	727,797 (2025)	8	2	0
Prague	Czechia	1,407,084 (2025)	7	3	0
Warsaw	Poland	1,863,800 (2024)	7	2	1
Singapore	Singapore	6,036,900 (2024)	7	3	0
Tokyo	Japan	9,983,365 (2026)	7	2	1
Total			112	37	11

Table 2: The sixteen cities distribute across the dimensions the indicators measure.

Dimension	Span within the set
Cost of living	Moderate capitals (Warsaw, Prague, Madrid, Berlin) to among the world's most expensive (Zurich, Oslo, London, Singapore)
Climate	Cold northern (Oslo, Helsinki, Stockholm) and continental (Berlin, Vienna, Warsaw, Prague) through maritime (London, Dublin, Amsterdam) and warm Mediterranean (Madrid, Milan) to tropical Singapore and humid-subtropical Tokyo
Scale	Mid-size capitals (Zurich, Oslo, Helsinki, Dublin) up to large metropolises (Tokyo, London, Paris)
Mobility	Cycling and tram networks (Amsterdam, Berlin, Prague, Vienna) to dense metro systems (Tokyo, Paris, London)
Safety	Low recorded homicide rates (Singapore, Tokyo, Oslo) alongside higher rates elsewhere in the set

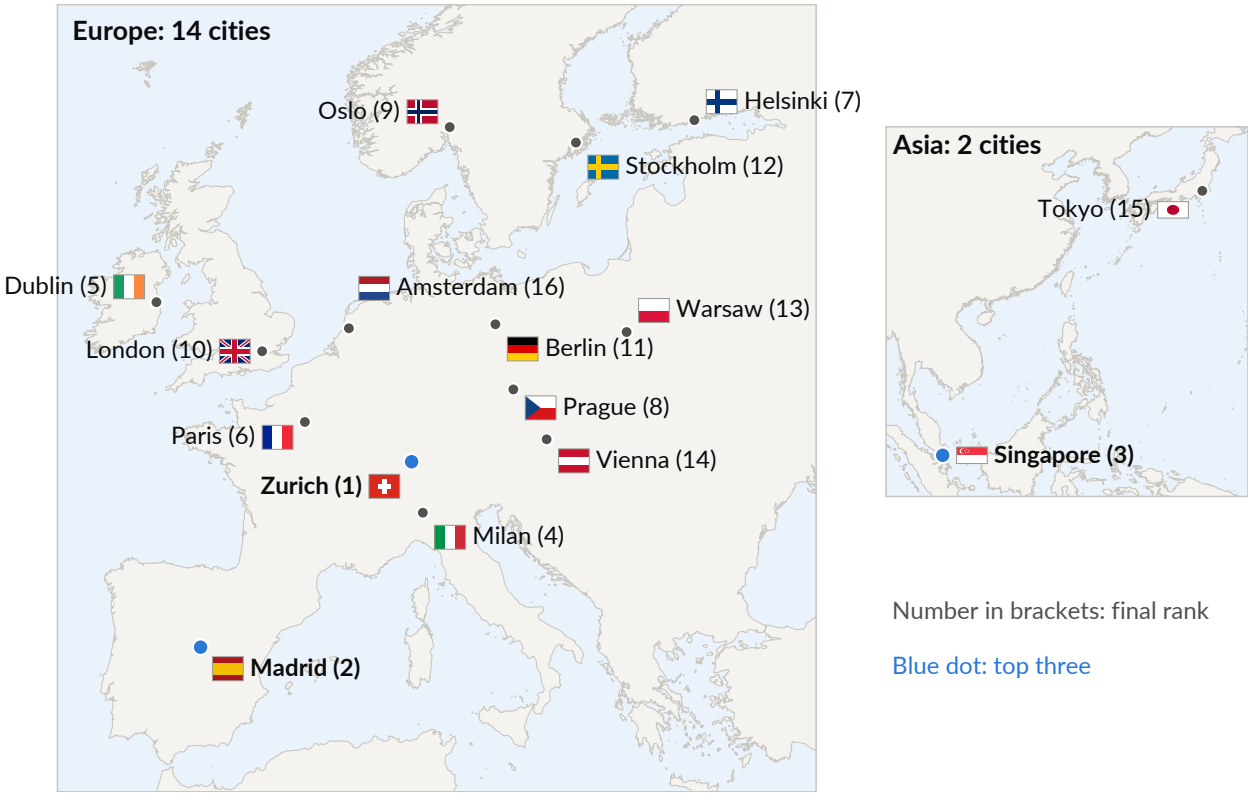


Figure 1: The 16 candidate cities, with national flags and final ranks in brackets; blue dots mark the top three.

4.2 Indicator system

The ten indicators form five categories, with two indicators in each category. Table 3 gives the unit, direction, anchor and data note in one place. Utilities lie on $[0, 1]$, with larger values always preferred.

For a benefit and cost indicator, respectively, utility is

$$u = \frac{x}{a + x}, \tag{1}$$

and

$$u = \frac{a}{a + x}. \tag{2}$$

The anchor a gives utility 0.5 when $x = a$. Higher-education density is $10^6 c/P$ institutions per million, with anchor 10 per million.

For transport stops and pools, facility counts become density per 100,000:

$$d = \frac{100000c}{P}. \tag{3}$$

We use $x = d$ in Equation 1; a verified subset gives the lower utility bound and the upper bound remains 1 when other providers are uncounted.

Table 3: Indicator definitions, anchors and data treatment.

Category	Indicator	Unit	Better when	Anchor a	Data note
Safety and environment	Homicide rate	Deaths per 100,000	Lower	2 deaths per 100,000	City record; Zurich recalculated
	$PM_{2.5}$ concentration	$\mu g/m^3$	Lower	15 $\mu g/m^3$	City record
Cost of living	Living-cost index	Index points	Lower	100 points	City record
	Housing price-to-income ratio	Ratio	Lower	10	City record
Education	Upper-secondary pupils per teacher	Pupils per teacher FTE	Lower	15 pupils/FTE	City record; proxy fills
	Higher-education institutions	Institutions per million	Higher	10 per million	City record
Mobility	Mean travel/commute time	Minutes	Lower	30-minute decay scale	City record; survey proxies and utility ranges
	Public-transport stops	Stops per 100,000	Higher	50 per 100,000	City record; utility ranges for uncounted providers
Daily life	Fixed broadband speed	Mbps	Higher	100 Mbps	City record
	Public swimming pools	Venues per 100,000	Higher	1 per 100,000	City record; utility ranges for uncounted providers

Upper-secondary teaching resources use pupils per full-time-equivalent (FTE) teacher:

$$x = \frac{\text{upper-secondary pupils}}{\text{upper-secondary teacher FTE}}. \tag{4}$$

Prague's 2024/25 records give about 11.6 pupils per teacher after lower-gymnasium years are removed [5].

For travel, a journey of t minutes has utility

$$q = \exp(-t/30). \tag{5}$$

When only a mean duration $\bar{t}$ is available, its utility range is

$$\mathbb{E}[q] \in [\exp(-\bar{t}/30), 1]. \tag{6}$$

The lower endpoint follows from convexity, and 1 is the upper limit for nonnegative durations. Work-commute means describe general travel conditions; Singapore's school-stage bins provide a separate range [12, 6].

Figure 3 compares the cities on the shared utility scale. Singapore and Milan have the highest homicide utilities (0.97 and 0.89), while Amsterdam has 0.06; Madrid has the lowest pupil-teacher proxy ratio at 9.6 pupils per teacher and utility 0.61.

Figure 2 shows the raw values, including Zurich's higher-education supply of 52 institutions per million.

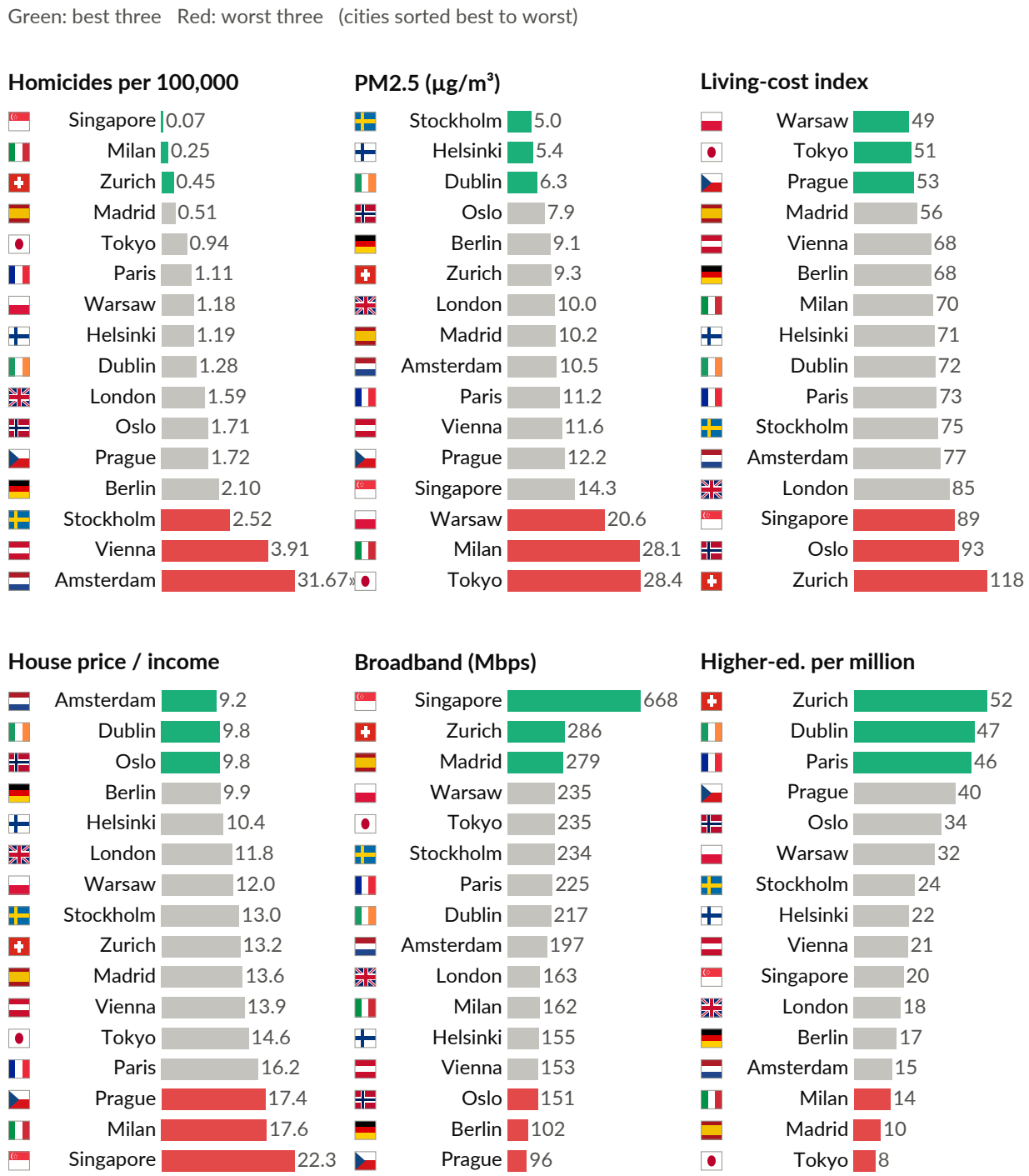


Figure 2: Raw values for the six indicators with complete city records, ordered from best to worst within each indicator. The best three cities are green and the worst three red; flags identify countries. Homicides are shown per 100,000, with Zurich's recalculated 0.45 and Amsterdam's 31.67 marked off-scale with "≫". Higher-education supply is institutions per million.

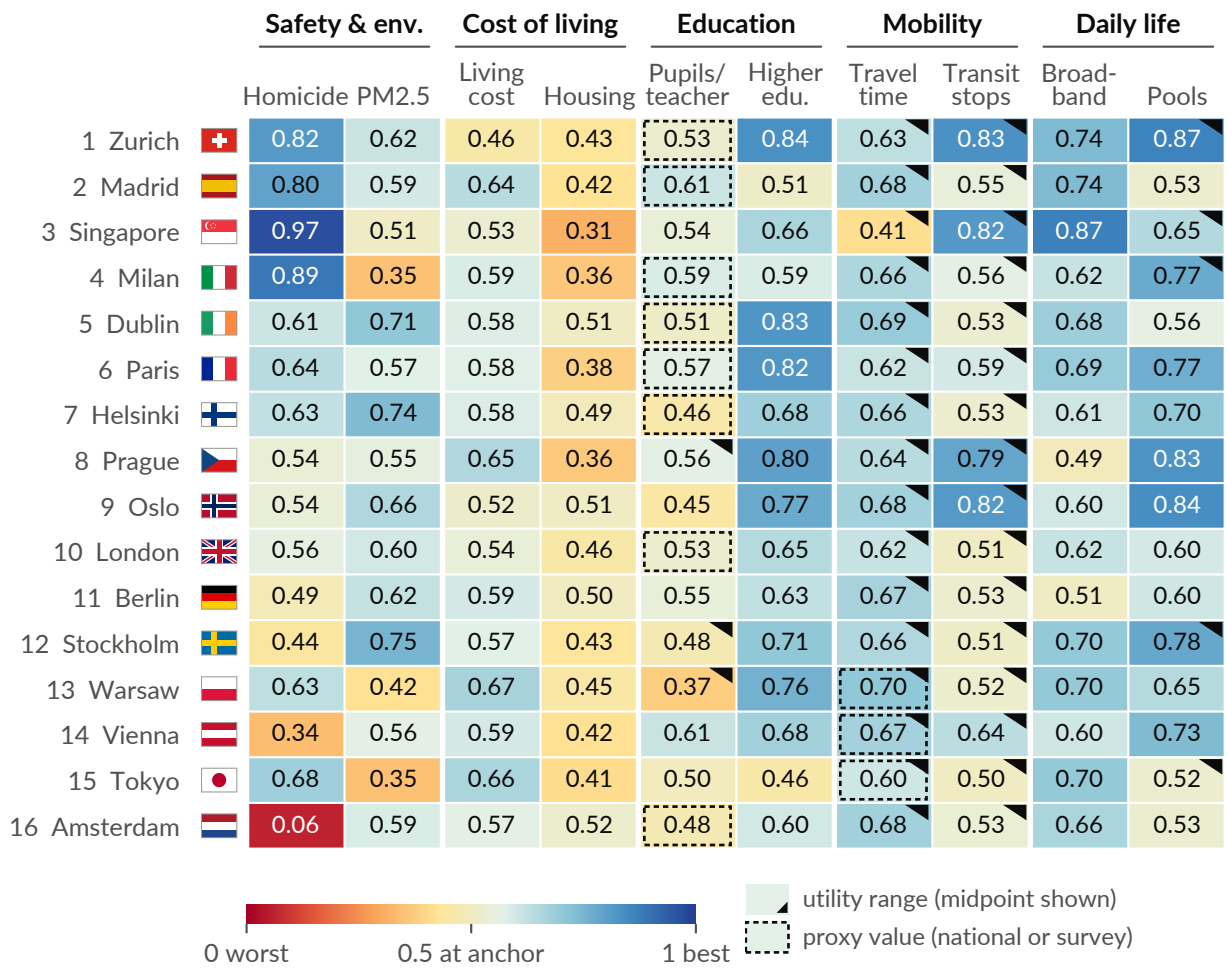


Figure 3: Indicator utilities for the sixteen cities, ordered by final score and grouped by category. Colour is utility from 0 (worst) to 1 (best). Utility ranges show their midpoint with a black corner triangle, proxies have a dashed outline, and flags mark countries.

4.3 Data sources and filled values

The team table supplies the base matrix and its source years and boundaries [1]. Table 4 lists the principal source families.

Three construction rules keep measurements comparable:

1. Count each station once, excluding separate platforms and repeated line entries. Warsaw's 21 M1 and 18 M2 entries give 38 stations after merging the shared Świętokrzyska interchange [14].

2. Match school stage before dividing pupils by teacher FTE. Prague's totals therefore remove lower-gymnasium pupils and teachers [5].

3. Recalculate Zurich's supplied 0.14 as 0.45 per 100,000: two completed homicide offences recorded by the Zurich city police in 2025, divided by 440,935 residents [11]. A rate built on two events is unstable.

Tokyo's Oedo directory contributes 38 distinct stations in the 23 wards, and Oslo's inventory uses parent stop groups; the questioned Tveten bru entry is excluded because its closure is unconfirmed [13, 4].

The 11 proxy cells are six Eurostat national pupil-teacher ratios from table educ_uoe_perp04 (ISCED 3, 2024), two median pupil-teacher values of 13.45 for Zurich and London, and three survey travel means entering as utility ranges: Vienna 32 minutes, Tokyo 47.5 minutes from the 2021 Japanese time-use survey, and Warsaw 27 minutes [9]. The Eurostat ratios are Helsinki 17.4, Amsterdam 16.4, Dublin 14.2, Paris 11.5, Milan 10.6 and Madrid 9.6.

The 160 entries comprise 123 single values, including proxies, and 37 utility ranges. Zurich's homicide recalculation is separate from the 11 proxies.

Table 4: Sources and years by indicator. Supplements establish counted subsets, proxies and boundary rules for the team matrix.

Indicator	Main sources	Year(s)	Notes
Homicide rate	Team table; Zurich police statistics [1, 11]; CBS recount for Amsterdam [10]	2022–2025	Zurich: two completed offences recorded by city police in 2025; 440,935 residents; rate 0.45 per 100,000.
$PM_{2.5}$	Team table	2021–2025	City monitoring values with reported station and year scopes.
Living-cost index	Team table [1]	2025–2026	City index values; source year stays attached to each row.
Housing price-to-income	Team table; Oslo property-price archive [1, 58]	2025–2026	City price-to-income values; administrative boundaries follow the team matrix.
Upper-secondary pupils per teacher	Team table; Prague, Paris, Finland and Ireland education records; Eurostat proxy fills [1, 5, 24, 54, 33, 53, 48, 23, 26, 27]	2024–2026	Six Eurostat national ratios and two median fills for Zurich and London are labelled proxies; Prague is stage-matched; Shanghai and Singapore stage tables support supplementary records.
Higher-education institutions	Team table; school and post-secondary directories [1, 40, 41]	2025–2026	Institutions per million; each provider counted once, with cross-stage providers flagged.
Mean travel/-commute time	Team table; Eurostat city series; Warsaw school-trip survey; Singapore census bins [1, 12, 9, 6, 25]	2011–2024	Vienna, Tokyo and Warsaw means enter as utility ranges; work-commute means are proxies for general travel conditions.
Public-transport stops	Operator and municipal inventories for Oslo, Zurich, Helsinki, Madrid, Vienna, Prague, Berlin, London, Singapore, Amsterdam, Milan, Stockholm, Tokyo and Warsaw [18, 4, 19, 20, 29, 34, 35, 36, 37, 38, 39, 42, 44, 55, 52, 51, 13, 14, 57, 30, 43, 56]	2024–2026	Operator inventories, counted once per station; interchange and boundary rules are explicit, with verified subsets forming lower bounds.
Fixed broadband speed	Team table	2025–2026	City speed observations with the reported test window.
Public swimming pools	Team table; municipal and operator inventories for Stockholm, Paris, Tokyo, Singapore, Zurich and Milan [1, 21, 22, 31, 32, 45, 46, 47, 49, 50]	2025–2026	Venues counted once per provider; operator inventories are lower bounds when coverage is incomplete.

5 Establishing the baseline evaluation model

5.1 Determining indicator weights (AHP)

Step 1: Set up the hierarchy. We use the analytic hierarchy process (AHP) to turn our student priorities into weights: the goal is city choice, followed by five categories and their ten indicators [2].

Step 2: Compare the five categories. Table 5 uses Saaty's 1–9 importance scale, with 1 for equal importance and reciprocal entries for reverse comparisons. Safety and schooling come first; cost and mobility support the family budget and daily journeys, while broadband and pools receive less weight.

Table 5: Pairwise category comparisons from our student viewpoint. S: safety and environment; C: cost of living; E: education; M: mobility; D: daily life.

Category	S	C	E	M	D
Safety and environment	1	3	1	3	4
Cost of living	1/3	1	1/3	1	2
Education	1	3	1	3	4
Mobility	1/3	1	1/3	1	2
Daily life	1/4	1/2	1/4	1/2	1

Step 3: Calculate category weights and check consistency. For comparison matrix A, find its largest eigenvalue $\lambda_{\max}$ and positive eigenvector v, then normalise v to sum to one for category weights g_k:

$$Av = \lambda_{\max} v. \tag{7}$$

With five categories ($n = 5$), the consistency index and consistency ratio are

$$CI = \frac{\lambda_{\max} - n}{n - 1}. \tag{8}$$

$$CR = \frac{CI}{RI}. \tag{9}$$

The random index is $RI = 1.12$; our matrix gives $\lambda_{\max} = 5.0264$, $CI = 0.00660$ and $CR = 0.0059 < 0.1$, meeting the consistency threshold [2].

Step 4: Compare the two indicators within each category. Our local importance ratios are 2:1 for homicide to $PM_{2.5}$, 1:1 for living costs to housing, 3:1 for teacher resources to higher-education supply, 2:1 for travel time to stops, and 2:1 for broadband to pools. These choices emphasise personal safety, current schooling, journey time and schoolwork; each pair and its reciprocal give an internally consistent two-indicator matrix. The local weight p_j is its share of the pair's ratio, giving global weight

$$w_j = g_k p_j. \tag{10}$$

Figure 4 shows both levels; calculations use unrounded weights.

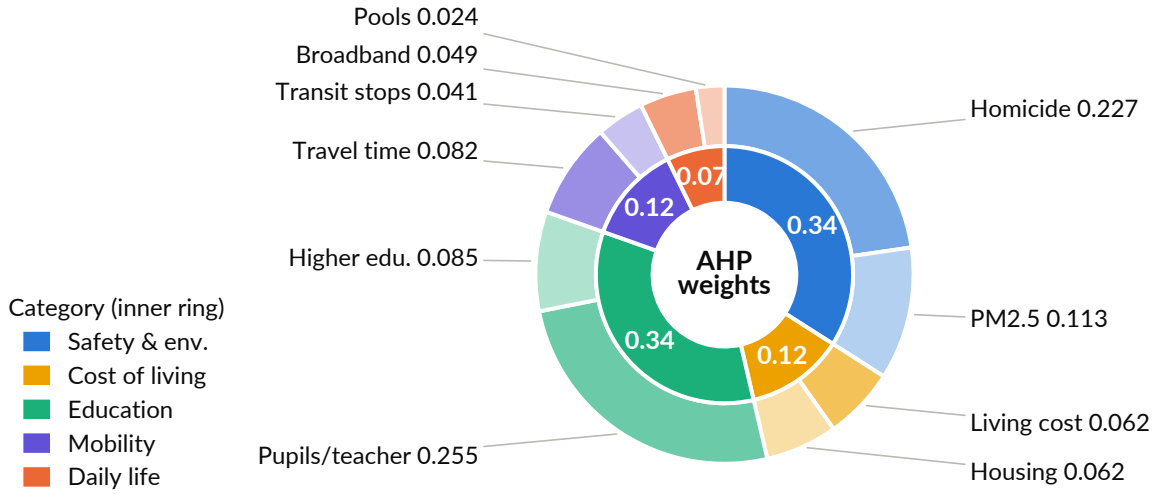


Figure 4: AHP weights from our student priorities. The inner ring shows the five category weights; the outer ring shows the ten global indicator weights, printed beside each slice. Safety and environment and education each receive 0.340.

5.2 TOPSIS scoring

Step 1: Form the utility matrix. For the representative score, using the arithmetic midpoint of each utility range $[\ell_{ij}, h_{ij}]$ is a modelling choice on the shared 0–1 scale. Endpoint values are used only in the robustness checks:

$$u_{ij} = \frac{\ell_{ij} + h_{ij}}{2}. \tag{11}$$

Step 2: Find the distance to the ideal. A smaller D_i^+ means fewer weighted shortfalls from the all-one ideal [3]:

$$D_i^+ = \sqrt{\sum_{j=1}^{10} w_j (1 - u_{ij})^2}. \tag{12}$$

Step 3: Find the distance to the worst profile. A larger D_i^- means stronger performance above the all-zero profile:

$$D_i^- = \sqrt{\sum_{j=1}^{10} w_j u_{ij}^2}. \tag{13}$$

Step 4: Calculate the score. TOPSIS gives the representative score S_i from 0 to 100, with higher values preferred:

$$S_i = 100\frac{D_i^-}{D_i^+ + D_i^-}. \tag{14}$$

Step 5: Carry utility ranges into the score. Since the score rises with every utility, the lower and upper utility vectors ℓ_i and h_i give the score bounds used in sensitivity checks:

$$[L_i, U_i] = [S(\ell_i), S(h_i)]. \tag{15}$$

5.3 Scoring results and interpretation

Zurich leads with a representative score of 64.5, followed by Madrid and Singapore (Table 6). Figure 5 contrasts tiers using representative cities: the whole top tier (ranks 1–3), the lower end of the middle tier (7–9) and the lowest three cities (14–16).

Table 6: Representative scores under AHP weights, using the midpoint of each utility range.

Rank	City	Score	Tier	Rank	City	Score	Tier
1	**Zurich**	**64.5**	**Top**	**9**	**Oslo**	57.6	Middle
2	**Madrid**	63.3	Top	**10**	**London**	56.1	Bottom
3	**Singapore**	62.5	Top	**11**	**Berlin**	55.6	Bottom
4	**Milan**	61.1	Middle	**12**	**Stockholm**	55.4	Bottom
5	**Dublin**	60.8	Middle	**13**	**Warsaw**	54.5	Bottom
6	**Paris**	60.8	Middle	**14**	**Vienna**	54.4	Bottom
7	**Helsinki**	58.7	Middle	**15**	**Tokyo**	54.0	Bottom
8	**Prague**	58.4	Middle	**16**	**Amsterdam**	45.2	Bottom

Top tier. Ranks 1–3, Zurich, Madrid and Singapore, score 62.5–64.5 through strength in safety and environment and education, the two heaviest categories. Zurich combines category scores of 0.75 and 0.61, Madrid 0.73 and 0.59, and Singapore 0.81 and 0.57.

Middle tier. Ranks 4–9, Milan through Oslo, score 57.6–61.1, starting 1.4 points below Singapore; a strong category is paired with a weaker one. For example, Helsinki scores 0.66 for safety and environment and 0.52 for education, while Prague scores 0.62 for education and 0.54 for safety and environment.

Bottom tier. Ranks 10–16, London through Amsterdam, score 45.2–56.1, starting 1.4 points below Oslo, as one or two heavily weighted utilities pull scores down. Vienna's safety and environment score is 0.41 despite education at 0.63; Amsterdam's homicide utility of 0.06 leaves it 8.8 points behind Tokyo.

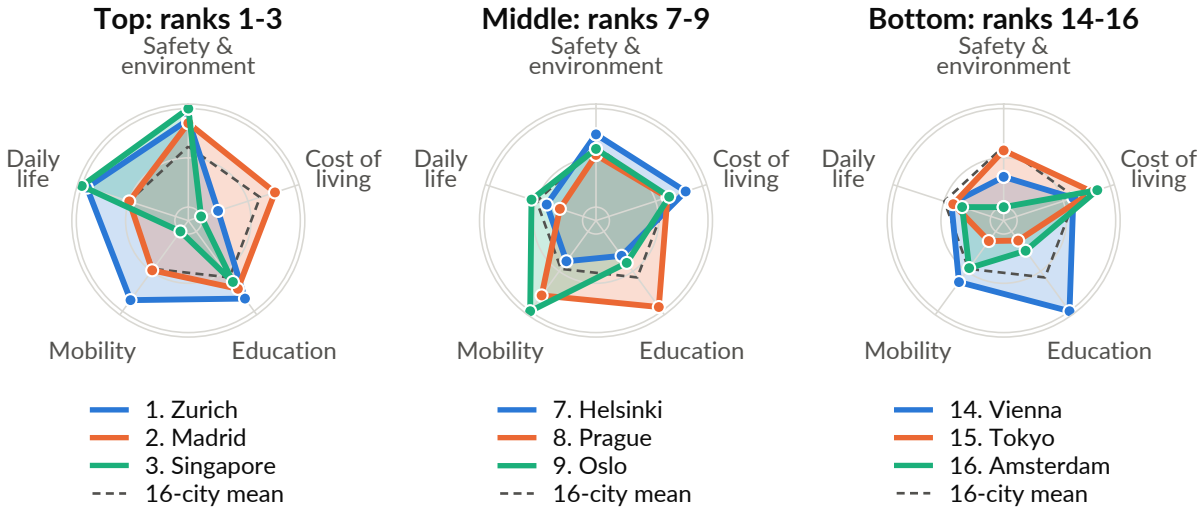


Figure 5: Representative cities chosen to contrast tiers: the whole top tier (ranks 1–3), the lower end of the middle tier (7–9), and the lowest three cities (14–16). Each axis is stretched from the lowest city (centre) to the highest city (outer ring) among all sixteen; the dashed line is the sixteen-city mean. Category scores combine each pair of indicators using its local AHP weights.

5.4 A significant revision: from equal weights to AHP weights

Equal weights give every indicator 0.1, making pools as influential as homicide or teacher resources. AHP expresses our student priorities: safety and environment and education each receive 0.340, cost and mobility 0.123 each, and daily life 0.073.

With utilities, anchors and scoring unchanged, the leading four shift from Zurich, Oslo, Paris and Dublin to Zurich, Madrid, Singapore and Milan. Twelve of sixteen cities change rank, with Kendall's $\tau = 0.517$ measuring agreement between the two rankings. Zurich stays first; the revised top group reflects the greater weight on safety and schooling.

5.5 Robustness of the recommendation

Score bounds and maximum regret. Figure 6 carries the score intervals, representative scores and weight-draw frequencies. Maximum regret is the most a city could trail the best rival when every utility range breaks against it:

$$R_i = \max\left\{0, \max_{k \neq i} U_k - L_i\right\}. \tag{16}$$

Zurich has the smallest maximum regret, 6.1 points; nine cities can be first under some endpoint combination.

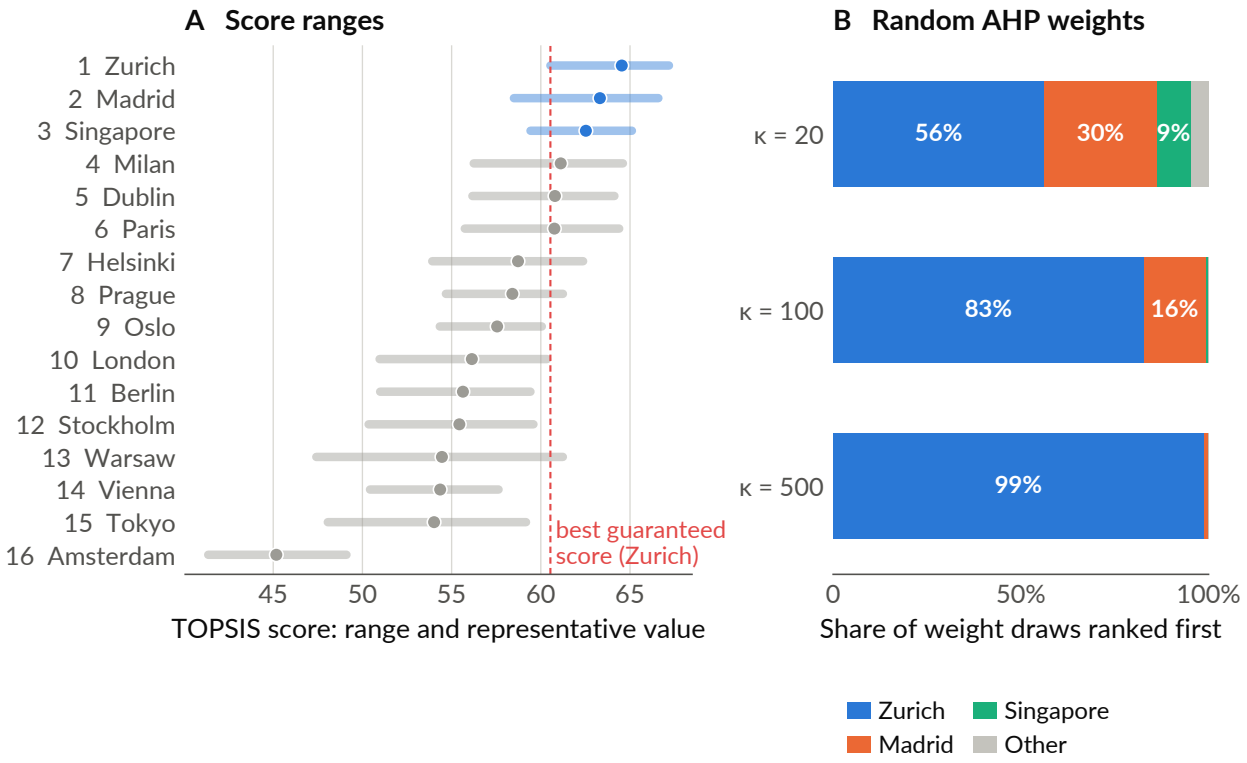


Figure 6: Robustness of the recommendation. Panel A shows each city's score interval as a light bar and its representative score as a dot; the top three are blue and the dashed line is Zurich's lower bound. Panel B shows the share of 10,000 AHP-centred Dirichlet weight draws in which Zurich, Madrid, Singapore or another city ranks first at concentrations 20, 100 and 500.

Dependence on homicide inputs. Zurich stays first up to about 0.69 per 100,000 (about three offences a year), where Madrid draws level. Withdrawing Zurich's homicide value (treating it as unknown, utility range $[0, 1]$) puts Zurich eighth and Madrid first. Amsterdam's supplied numerator is not reproduced by official CBS counts; withdrawing its homicide value under the same rule lifts Amsterdam to fifteenth [10].

Data withdrawals. Replacing selected point cells with the full utility range $[0, 1]$ changes the leading city as shown in Table 7.

Table 7: First city after withdrawing dated or flagged point cells.

Withdrawn group	Cells	First (AHP)	First (equal weights)
Before 2020	13	Zurich	Zurich
After 2025	64	Madrid	Zurich
Flagged entries	28	Milan	Zurich
Union of the three groups	95	Milan	Milan

Source choices. Zurich leads under both AHP and equal weights in the baseline. All 31 source-variant checks in Table 8 keep Zurich first under AHP; these cases vary counting rules and sources separately from the broader withdrawals in Table 7.

Table 8: The 31 source-variant cases in nine groups, including the before-update and current comparison cases.

Group	Cases	What changes	First city
Before-update baseline	1	Baseline before the source refinements	Zurich
Current comparison	1	Conditional source refinements included	Zurich
Prague bus inventory	2	Withdraw refinement; withhold request-stop witnesses	Zurich
Vienna bus inventory	11	Withdraw refinement; withhold selected route witnesses in nine single-route cases and one nine-route case	Zurich
Oslo bus inventory	5	Withdraw refinement; withhold two rail-replacement stops, Midtstuen, or all three service flags; include questioned Tveten bru stop	Zurich
Singapore pools	5	Withdraw OCBC, directory refinement, or all supplements; withhold three or four notice venues	Zurich
Zurich pools	1	Withdraw refinement	Zurich
Metro inventories	4	Withdraw Tokyo, Warsaw, both, or Amsterdam mapped inventory	Zurich
Travel	1	Withdraw Singapore historical school-stage proxy	Zurich
Total	31		Zurich

Priority checks. In 10,000 draws around the AHP weights, $w \sim \mathrm{Dirichlet}(\kappa w_{\mathrm{AHP}})$, Zurich leads in 56.3%, 82.8% and 98.7% at concentrations $\kappa = 20, 100, 500$. Larger κ keeps

priorities closer to AHP. Doubling education or mobility weights separately also keeps Zurich first.

Compromise ranking. VIKOR uses the same utilities and weights. It scales each indicator's shortfall between the best and worst observed utility, then equally combines the normalized weighted total and largest shortfall [17]. Madrid, Zurich and Paris lead. Stability holds, but Madrid's lead is below the required $1/(16-1)$ advantage; Madrid and Zurich remain close alternatives.

6 Model revision and extension

6.1 Looking ahead to 2035

We test more public services (S1), higher costs and congestion (S2), and smaller classes, cleaner air and faster travel (S3) as possible 2035 scenarios. S1 increases higher-education institutions, transit stops and pools by 10%; S2 increases living costs, housing price-to-income and travel time by 10%; S3 reduces pupils per teacher, $PM_{2.5}$ and travel time by 10%. These hypothetical changes use the same formulas, anchors and weights in every city.

Figure 7 shows the top six: Zurich, Madrid and Singapore stay first, second and third, and Dublin and Paris swap fifth and sixth under S3. Ranks 7–16 do not change in any scenario. More services add 0.1–0.3 points, higher costs and congestion subtract 0.4–0.5, and smaller classes with cleaner air and faster travel add about 1.0.

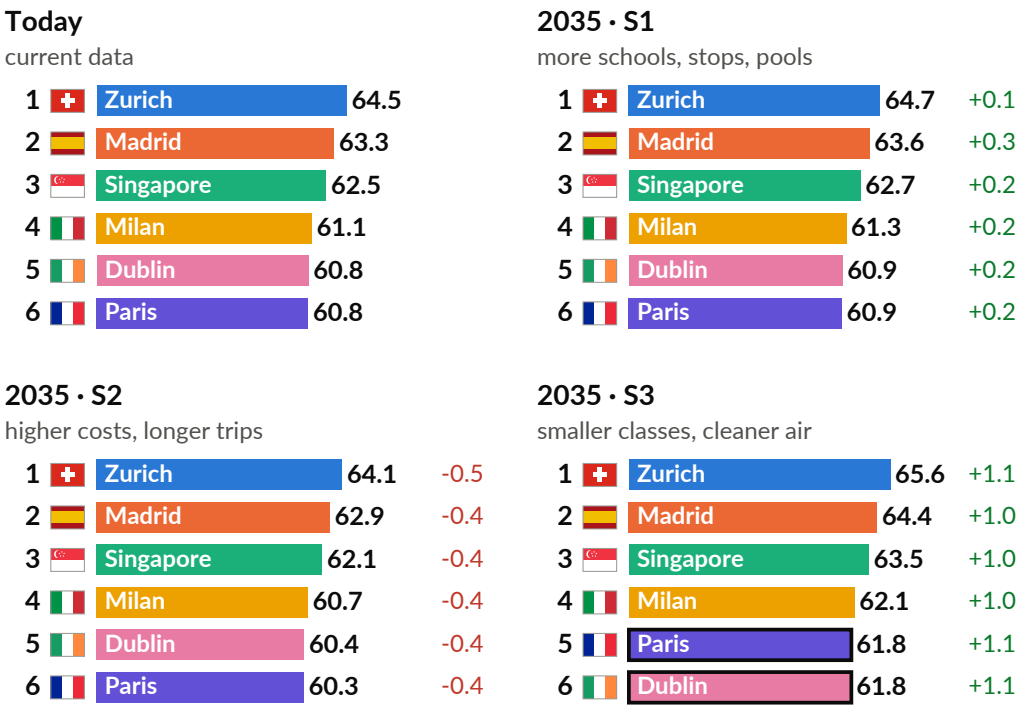


Figure 7: Top six cities today and under three hypothetical 2035 scenarios. Bars show representative scores; numbers give changes from today, and boxes mark changed ranks. S1: 10% more higher-education institutions, transit stops and pools. S2: 10% higher living costs, housing price-to-income and travel time. S3: 10% fewer pupils per teacher, lower $PM_{2.5}$ and shorter travel time.

6.2 Within-city fairness and growth stages

Citywide supply can mask differences in access and age. Figure 8 shows walking-time context from resident grids [15]. Istanbul's 2012 cohort (ages 14–18) has a minimum travel-utility gap of 0.227 between all students and the lowest fifth [28]. In Oslo's 2023 survey, 66% of grade 8 and 41% of final-year upper-secondary respondents rated youth meeting places positively [16].

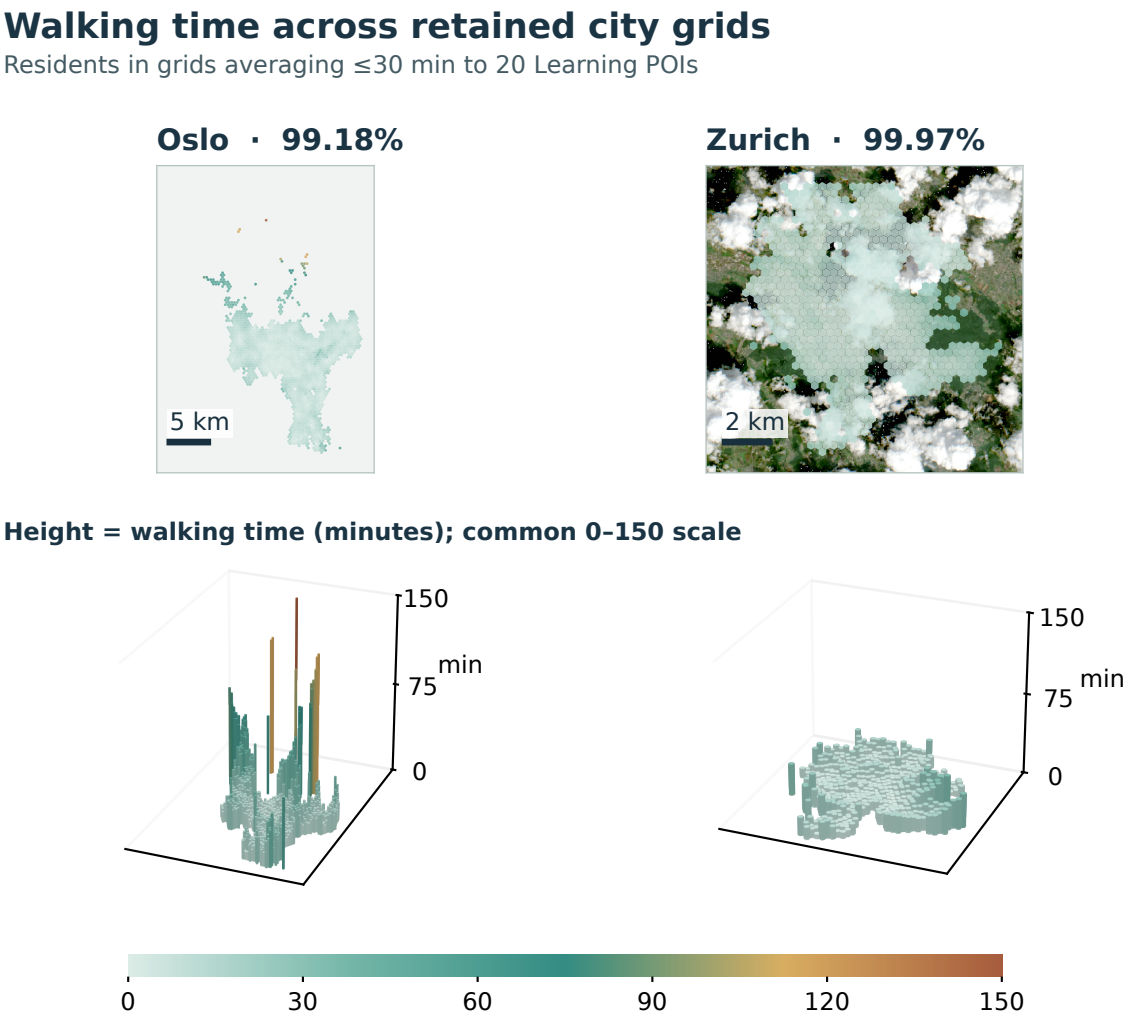


Figure 8: Walking-time context from the brunogrids resident grids. The modelled mean walking time is measured from resident grid cells to 20 learning POIs, with one common 0–150 minute scale. Among retained residents, 99.18% of Oslo and 99.97% of Zurich fall within 30 minutes. This context uses the source's resident grids and is not part of the ten-indicator score.

7 Conclusion and decision use

Zurich is our choice among these sixteen cities under the team's high-schooler priorities, supported by strong safety and education scores. It leads under AHP and equal weights, in all 31 source-variant checks, and in 82.8% of 10,000 weight trials at $\kappa = 100$, with the smallest maximum regret of 6.1 points. Madrid is 1.2 points behind and leads when Zurich's homicide value is withdrawn (treated as unknown, utility range $[0, 1]$). Broad withdrawals of dated or flagged entries can also make Madrid or Milan first (Table 7). All three hypothetical 2035 scenarios retain Zurich, Madrid and Singapore as the top three. Families can use this shortlist, then check school options, admission, fees and daily routes against the student's needs.

A city choice built around a student's day

Zurich is our first choice among sixteen cities for a high schooler, with 64.5 points. Madrid follows just 1.2 points behind, and Singapore takes third. The result gives families a starting point for comparing places to live and study.

What matters during a school day?

We scored each city on ten everyday measures and weighted them by what matters most to a teenager. These cover safety, air quality, living and housing costs, teaching resources, higher education, transport stops, travel time, broadband and swimming facilities. Our team chose the priorities, giving safety and education the greatest influence.

Zurich combines strong school resources and safety with transport and daily facilities that help balance its higher costs. It stays first in all 31 comparisons using different counting rules and sources for stops, pools and travel. Treating all 64 entries dated after 2025 as unknown puts Madrid first; doing so for 28 entries flagged for data concerns, or all 95 entries in the combined date-and-quality check, puts Milan first. Zurich leads in 82.8% of 10,000 alternative priority choices clustered around our team's priorities. When uncertain evidence works against each city, Zurich falls fewer points behind the best rival than any other choice.

Why Madrid deserves a close look

Zurich's lead depends on its homicide figure. We use 0.45 per 100,000, calculated from two completed homicide offences recorded by the Zurich city police in 2025 and 440,935 residents [11]. A rate based on two events is unstable. Zurich stays first up to about 0.69 per 100,000, about three offences a year. Treating its homicide figure as unknown puts Madrid first.

Eleven entries use substitutes: six national pupil-teacher ratios from Eurostat, two median ratios for Zurich and London, and three survey travel means. The travel means allow a span of possible scores. Zurich's homicide recalculation is separate from these substitutes.

Making the choice personal

Families should check school admission, language, friendships and wellbeing alongside the scores. A stop count needs a suitable route and timetable; a pool count needs affordable fees and usable opening hours. Visiting schools and trying the daily journey can help settle a close choice.

Zurich remains first when we double the importance of schooling or travel separately. Three possible 2035 scenarios also keep Zurich, Madrid and Singapore at the top, with Dublin and Paris switching places in one scenario. These tests show how the scores respond to changes in services, costs, schools and air quality.

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Report on Use of AI Tools

The team used AI-assisted retrieval, calculation and drafting during 8–14 November 2026. We checked outputs against official source records, saved retrieval receipts, recomputed model results, and compiled and reviewed the rendered manuscript. The tools below record the uses documented in the team report.[7, 8]

Table 9: Recorded AI-tool use and checks.

Tool	Who/when (as recorded)	Used for	How we checked it
OpenAI Codex	Team record, 8–11 and 14 November 2026	Official-source search and extraction, data-scope checks, Python calculations and audits, LaTeX drafting, wording revisions, and simulated judge reviews of the draft	Compared claims with source records; recomputed calculations; compiled and reviewed the rendered pages
Anthropic Claude (Claude Code)	14 November 2026	Planning the revision after our advisor's feedback, AHP and scenario calculations, all figures, proxy-data searches, and small text edits	Recomputed every reported number from the result files; inspected each figure at print size
Exa (`web_fetch_exa`)	Team record, 8–9 November 2026	Retrieval and text extraction from official webpages, PDFs and data interfaces	Saved source URLs and returned text; checked extracted representations against original downloaded records and excluded ambiguous material
Anti-defensive-writing skill	Applied during manuscript revision	Editorial guidance for concise, evidence-supported wording	Reviewed claims against the manuscript evidence and retained conclusion-changing checks

Usage by other team members: to be added by the team before submission.